

GRAND RAPIDS, MN, USA

WMO: 727458

Lat: 47.211N

Lon: 93.510W

Elev: 413

StdP: 96.46

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.7	-26.2	-32.7	0.2	-26.6	-31.1	0.2	-24.7	11.1	-11.8	10.2	-11.4	2.5	310	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	29.0	21.0	27.6	19.8	26.3	18.7	22.5	27.5	21.2	26.2	20.0	24.8	4.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.4	26.4	19.1	14.6	24.2	18.0	13.6	22.9	68.3	27.6	63.4	26.3	59.3	24.7	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.7	8.6	7.7	-31.9	31.9	3.4	1.5	-34.3	33.0	-36.3	33.9	-38.2	34.7	-40.6	35.8
			-31.3	24.4	2.4	1.2	-33.0	25.2	-34.4	25.9	-35.8	26.6	-37.5	27.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-12.5	-10.2	-2.9	4.3	11.2	16.7	19.5	18.2	13.9	6.7	-1.9	-9.1
	DBStd	12.41	7.58	7.25	7.01	5.29	4.51	3.64	3.11	3.07	4.57	4.83	5.99	6.89
	HDD10.0	2978	697	565	403	180	38	2	0	0	16	128	358	592
	HDD18.3	5161	956	798	659	419	226	71	24	41	146	362	607	851
	CDD10.0	993	0	0	2	11	76	203	294	253	131	24	1	0
	CDD18.3	134	0	0	0	0	4	23	59	36	12	1	0	0
	CDH23.3	1209	0	0	0	4	73	235	516	282	92	7	0	0
	CDH26.7	246	0	0	0	0	13	49	118	51	14	1	0	0
Wind	WSAvg	3.4	3.6	3.6	3.5	3.6	3.5	3.0	2.9	2.8	3.3	3.6	3.7	3.5
Precipitation	PrecAvg	697	23	13	32	54	80	103	100	91	83	59	33	25
	PrecMax	965	80	38	76	108	163	191	240	255	191	173	80	70
	PrecMin	457	2	1	4	8	10	13	20	16	14	10	3	5
	PrecStd	115	18	9	19	29	35	44	48	61	47	45	23	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	7.8	17.5	22.8	28.4	30.9	32.1	31.0	28.0	23.9	15.9	6.3
		MCWB	2.1	3.7	11.1	13.5	18.4	22.3	23.6	22.4	20.2	15.4	10.9	2.8
	2%	DB	2.1	4.1	12.4	19.0	25.9	27.8	29.1	27.8	26.2	20.0	12.1	3.3
		MCWB	0.3	1.4	7.2	10.4	15.9	19.6	21.5	20.2	18.9	13.8	7.9	1.1
	5%	DB	-0.1	2.4	8.9	17.1	22.7	26.2	27.7	26.4	23.7	17.2	8.5	1.4
		MCWB	-1.2	0.3	5.2	8.6	14.1	18.3	20.3	19.0	17.7	11.8	5.3	-0.2
	10%	DB	-2.2	0.6	6.3	13.8	20.1	24.0	26.3	24.8	22.0	14.0	6.3	0.1
		MCWB	-3.4	-1.2	2.8	6.9	12.6	16.9	19.3	18.3	16.6	9.7	3.4	-0.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	4.3	13.3	14.4	20.0	23.3	24.5	23.2	21.8	16.7	11.5	3.5
		MCDB	5.1	7.4	16.6	22.0	26.0	28.7	29.8	28.4	26.5	21.6	14.7	5.3
	2%	WB	0.3	2.0	7.8	11.1	17.4	21.1	22.9	21.7	20.0	14.5	7.8	1.2
		MCDB	1.7	4.2	11.3	17.5	22.9	26.8	28.0	26.7	24.5	19.0	10.5	3.2
	5%	WB	-1.3	0.2	5.3	9.2	15.2	19.7	21.5	20.3	18.7	12.4	5.5	0.1
		MCDB	-0.1	2.4	8.8	15.9	20.6	24.7	26.4	24.7	22.8	16.3	8.4	1.4
	10%	WB	-3.2	-1.4	3.1	7.4	13.5	18.1	20.3	19.2	17.2	10.3	3.6	-1.2
		MCDB	-2.2	0.3	6.2	13.1	19.1	22.7	25.0	23.5	20.8	13.8	6.2	0.0
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.5	10.4	11.2	12.1	11.4	11.2	11.0	10.7	8.8	7.7	8.0
		MCDBR	9.6	10.5	12.9	16.5	16.2	14.0	12.8	12.8	13.2	13.4	11.2	9.0
		MCWBR	8.2	8.1	8.7	9.9	8.8	7.7	6.9	6.8	7.7	8.2	8.0	7.5
	5% WB	MCDBR	9.1	9.8	12.0	14.8	13.5	12.1	11.8	11.2	11.2	11.9	10.3	8.3
		MCWBR	7.9	7.7	8.4	9.5	8.1	7.7	7.1	6.7	7.1	8.2	7.8	7.1
Clear-Sky Solar Irradiance	taub	0.251	0.268	0.306	0.348	0.364	0.369	0.379	0.375	0.349	0.307	0.284	0.263	
	taud	2.377	2.334	2.381	2.373	2.380	2.389	2.379	2.416	2.466	2.574	2.520	2.402	
	Ebn at Noon	855	909	920	907	900	894	879	869	863	860	812	791	
	Edh at Noon	68	89	101	112	116	116	115	107	93	71	61	60	
All-Sky Solar Radiation	RadAvg	1.46	2.59	3.74	4.65	5.27	5.66	6.07	5.12	3.82	2.29	1.35	1.03	
	RadStd	0.17	0.24	0.35	0.46	0.39	0.42	0.26	0.39	0.29	0.31	0.16	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	-1.54	N/A	N/A	N/A	+0.93	N/A	N/A	N/A	N/A

Nomenclature: See separate page